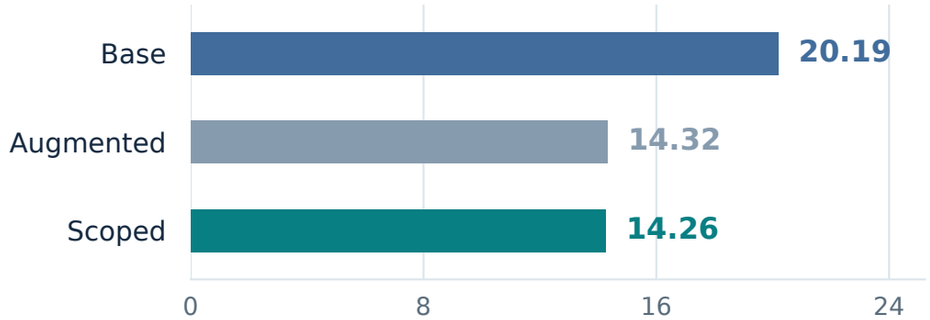
Targeted examples improve point estimates

Fixed 217 features · 64 new training circuits · two separately frozen synthetic cohorts · lower MAE is better

A First prospective check · 40 circuits

Mean absolute LUT4 error · three frozen point predictors

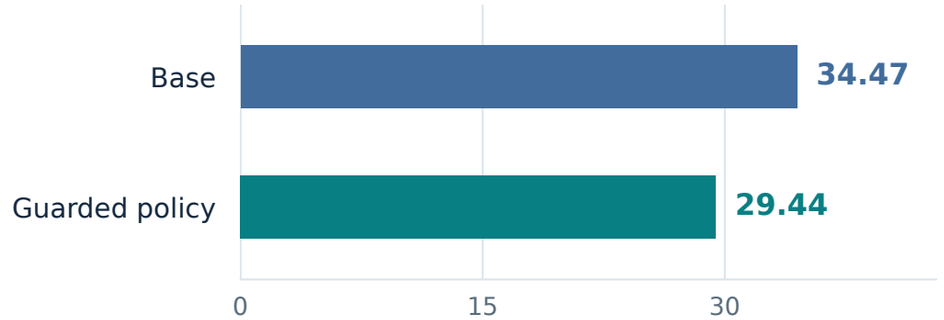


MAPE: 17.23% → 14.54% / 14.56% (augmented / scoped)

Fresh gate passed; retained development budget gate failed.

B New policy check · 32 other circuits

Mean absolute LUT4 error · zero additional fits



MAPE: 20.64% → 18.01% · MAE -14.58%

Development and new fresh gates passed; explicit experiment only.

Unchanged fresh counts · base → scoped (augmented also identical)

Cap LUT4	False accept	False reject
32	**0 → 0**	**4 → 4**
64	**0 → 0**	**0 → 0**
128	**5 → 5**	**0 → 0**

Worst underprediction: 83.53 → 82.83 LUT4

Unchanged fresh counts · base → guarded policy

Cap LUT4	False accept	False reject
32	**0 → 0**	**4 → 4**
64	**5 → 5**	**0 → 0**
128	**3 → 3**	**0 → 0**

Worst underprediction: 86.80 → 86.80 LUT4

Guarded policy: p = min(max(b, s), nextcap(b))

b = base; s = scoped; nextcap = smallest 32/64/128 ≥ b, else ∞.

The policy exactly preserves base decisions at these caps, including existing mistakes, and never lowers a base point estimate.

Regressions: retained 32-case stress MAE 15.52 → 15.67 / MAPE 13.42% → 13.78%; new four-case signed dual arithmetic MAE 18.04 → 20.63.

First 40-case test is consumed development for the later policy. Separate cohorts; no learning curve. Base remains the default.

Synthetic circuit evidence only; no calibrated safety, application-transfer, timing, power or full-CPU guarantee.